

Gelei Deng

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Profile

Ph.D. in Cybersecurity with extensive experience in AI security, software security auditing, and penetration testing. Skilled in Python, Solidity, and other mainstream programming languages. Passionate about advancing security technologies and methodologies and committed to bridging academic research with industry applications.

Areas of Expertise

- AI Security and System Security
- Blockchain Security
- Penetration Testing and Software Security Testing.
- Proficient in Python and Solidity; Familiar with other mainstream Languages including JavaScript, PHP, Rust, C#, etc.

Education

Nanyang Technological University (Aug 2020 – Oct 2024)

- Ph.D. Computer Science, Cybersecurity
- NTU Cyber Security Lab
- Main Research Interests on System Security, Web Security and Penetration Testing

Singapore University of Technology and Design (May 2015 – Sep 2018)

- Department of Engineering Product Design, B.E. Electrical Engineering
- Singapore Ministry of Education Full Scholarship (SM2) Holder
- Engineering Product Design Track, Honor List

Employment

Nanyang Technological University (Dec 2024 - Present)

Research Assistant Professor

- Conduct pioneering research on AI security, with focus on LLM security and red teaming methodologies at NTU Digital Trust Center
- Lead academic initiatives in developing novel threat detection and mitigation techniques

Quantstamp, Inc. (Oct 2022 – Present)

Lead AI Engineer, Auditing Engineer

- Conduct comprehensive security audits for blockchain projects (DeFi, wallets, DEXs, NFTs).
- Lead R&D on leveraging Large Language Models for automated smart contract auditing, fuzzing, and LLM agents' development.

OpenAI (Jan 2024 – Present)

Independent Contractor at OpenAI Red Teaming Network

- Participate in OpenAI-led red teaming to assess risks and safety profiles of OpenAI models and systems (e.g., GPT4o, DALL-E3, Operator).

Institute for Infocomm Research (I²R, A*STAR) (Jan 2019 – Jul 2020)

- Research Engineer & Penetration Tester.
- Performed penetration testing and research for Singapore government agencies.

Academic Research

Cyber Security Lab, NTU (July 2020 - Present)

- Ph.D. student in Cybersecurity
- Main research topics include system security, robotics security, Web3 security, and penetration testing automation.
- Collaborate with industrial partners (Huawei, etc.) to verify system security and stability.

SUTD-MIT International Design Center (2016-2018)

- Work on hardware-oriented digital signal processing
- Leading the research work in IIR filter design
- Involved in IDCT image processing research work

Publications

A complete list of publication: <https://scholar.google.com/citations?user=NuzUkMEAAAAJ>
(As to March 2025) Citation: 2054, h-index:19

First-Author Publications

Gelei Deng, Yi Liu, Víctor Mayoral-Vilches, Peng Liu, Yuekang Li, Yuan Xu, Tianwei Zhang, Yang Liu, Martin Pinzger, Stefan Rass, “PentestGPT: An LLM-empowered Automatic Penetration Testing Tool,” in *33rd USENIX Security Symposium (USENIX '24)*. Distinguished Artifact Award, 2024. Citation: 114.

Gelei Deng, Yi Liu, Kailong Wang, Yuekang Li, Tianwei Zhang, Yang Liu, “PANDORA: Jailbreak GPTs by Retrieval Augmented Generation Poisoning,” in *Workshop on Artificial Intelligence System with Confidential Computing (AISCC)*, Distinguished Paper Award, February, 2024. Citation: 37.

Gelei Deng, Yi Liu, Yuekang Li, Kailong Wang, Ying Zhang, Zefeng Li, Haoyu Wang, Tianwei Zhang, Yang Liu, “MASTERKEY: Automated Jailbreaking of Large Language Model Chatbots,” in *Network and Distributed System Security Symposium (NDSS '24)*. 2024. Citation: 334.

Gelei Deng, Zhiyi Zhang, Yuekang Li, Yi Liu, Tianwei Zhang, Yang Liu, Guo Yu, Dongjin Wang, “NAUTILUS: Automated RESTful API Vulnerability Detection,” in *32nd USENIX Security Symposium (USENIX '23)*. 2023. Citation: 19

Gelei Deng, Guowen Xu, Yuan Zhou, Tianwei Zhang, and Yang Liu, “On the (In)Security of Secure ROS2,” in *Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security (CCS '22)*. Association for Computing Machinery, New York, NY, USA, 739–753. Citation: 33.

Gelei Deng, Yuan Zhou, Yuan Xu, Tianwei Zhang, and Yang Liu, “An Investigation of Byzantine Threats in Multi-Robot Systems,” in *24th International Symposium on Research in Attacks, Intrusions and Defenses (RAID '21)*, New York, NY, USA, 17–32. Citation: 29

Gelei Deng, Stefanie Yu Xingjie and Huaqun Guo, “Efficient Password Guessing based on a Password Segmentation Approach,” in *GLOBECOM 2019*, Waikoloa, Hawaii, 2019. Citation: 16

Other Selected Publications

Zihao Xu, Yi Liu, **Gelei Deng**, Yuekang Li, and Stjepan Pice, “A Comprehensive Study of Jailbreak Attack versus Defense for Large Language Models,” in *Findings of the Association for Computational Linguistics ACL*, 2024.

Yuxi Li, Yi Liu, **Gelei Deng**, Ying Zhang, Wenjia Song, Ling Shi, Kailong Wang, Yuekang Li, Yang Liu, and Haoyu Wang, “Glitch Tokens in Large Language Models: Categorization Taxonomy and Effective Detection,” in *Proceedings of ACM Software Engineering*, FSE, Article 92, July 2024.

Xingshuo Han, Haozhao Wang, Kangqiao Zhao, **Gelei Deng**, Yuan Xu, Hangcheng Liu, Han Qiu, Tianwei Zhang, “VisionGuard: Secure and Robust Visual Perception of Autonomous Vehicles in Practice,” in *ACM Conference on Computer and Communications Security (CCS)*, October, 2024.

Yuan Xu, **Gelei Deng**, Xingshuo Han, Guanlin Li, Han Qiu, Tianwei Zhang, “PhyScout: Detecting Sensor Spoofing Attacks via Spatio-temporal Consistency,” in *ACM Conference on Computer and Communications Security (CCS)*, October, 2024.

Kunsheng Tang, Wenbo Zhou, Jie Zhang, Aishan Liu, **Gelei Deng**, Shuai Li, Peigui Qi, Weiming Zhang, Tianwei Zhang, Nenghai Yu, “GenderCARE: A Comprehensive Framework for Assessing and Reducing Gender Bias in Large Language Models,” in *ACM Conference on Computer and Communications Security (CCS)*, October, 2024.

Yi Liu*, **Gelei Deng***, Zhengzi Xu, Yuekang Li, Yaowen Zheng, Ying Zhang, Lidao Zhao, Tianwei Zhang, Yang Liu, “Jailbreaking ChatGPT via Prompt Engineering: An Empirical Study,” in *Proceedings of the 4th International Workshop on Software Engineering and AI for Data Quality in Cyber-Physical Systems/Internet of Things*. 2024

Yi Liu*, **Gelei Deng***, Yuekang Li, Kailong Wang, Zihao Wang, Xiaofeng Wang, Tianwei Zhang, Yepang Liu, Haoyu Wang, Yan Zheng, Yang Liu, “Prompt Injection Attack Against LLM-Integrated Applications,” Arxiv. 2023.

Ruichao Liang, Jing Chen, Kun He, Yueming Wu, **Gelei Deng**, Ruiying Du, Cong Wu, “PonziGuard: Detecting Ponzi Schemes on Ethereum with Contract Runtime Behavior Graph (CRBG),” in *Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (ICSE 2023)*, 2023

Yi Liu, Yuekang Li, **Gelei Deng**, Felix Juefei-Xu, Yao Du, Cen Zhang, Chengwei Liu, Yeting Li, Lei Ma, Yang Liu, “ASTER: Automatic Speech Recognition System Accessibility Testing for Stutterers,” in *38th IEEE/ACM International Conference on Automated Software Engineering (ASE 2023)*. 2023.

Yuan Xu, Xingshuo Han, **Gelei Deng**, Jiwei Li, Tianwei Zhang, Yang Liu, “SoK: Rethinking Sensor Spoofing Attacks against Robotic Vehicles from a Systematic View,” in *8th IEEE European Symposium on Security and Privacy (Euro S&P 23)*. 2023.

Yisroel Mirsky, Ambra Demontis, Jaidip Kotak, Ram Shankar, **Deng Gelei**, Liu Yang, Xiangyu Zhang, Maura Pintor, Wenke Lee, Yuval Elovici, Battista Biggio, “The Threat of Offensive AI to Organizations,” *Computer & Security*, vol.124, 2023.

Yi Liu, Yuekang Li, **Gelei Deng**, Yang Liu, et al., "Morest: Model-based RESTful API Testing with Execution Feedback," in *2022 IEEE/ACM 44th International Conference on Software Engineering (ICSE)*, 2022.

Yuan Xu, **Gelei Deng**, Tianwei Zhang, Han Qiu, Yungang Bao, "Novel denial-of-service attacks against cloud-based multi-robot systems," *Information Sciences*, Volume 576, 2021, Pages 329-344.

Luying Zhou, Huaqun Guo and **Gelei Deng**, "A fog computing based approach to DDoS mitigation in IIoT systems," *Computer & Security*, vol. 85, pp. 51-62, 2019.

Certificates, Awards and Activities

Projects

PentestGPT: An LLM-empowered Automatic Penetration Testing Tool

- Open-source tool with 8k stars on GitHub: <http://pentest-gpt.com/>
- Various collaboration with industrial partners (Huawei, ByteDance, etc.)
- Active open-source community with contributors

CVEs and Vulnerability Identification

More than verified 10 CVEs including: CVE-2021-39114, CVE-2021-30224, CVE-2021-37392, CVE-2021-37393, CVE-2021-37476, CVE-2021-37477

Multiple vulnerabilities and bugs confirmed by vendors including: Atlassian Confluence, Apache Magento, Bitbucket, SEO Panel, Spree Commerce, etc.

Certificates

- Offensive Security Web Expert (OSWE) (2021)
- Offensive Security Certified Professional (OSCP) (2020)
- BlackHat Advanced Infrastructure Hacking Completion (2019)

Awards

SUTD Y2015 Student Honor List (2018)

- Awarded to be one of the students with best overall performance and grade
- 12 out of more than 400

The Most Amazing SUTD Student Works of the Year (2018) – Drone ranger

- 5 selected projects out of more 100 student projects in 2018

Additional Information

- Fluent in English and Mandarin (spoken & written)
- Actively participating varieties of sports activities